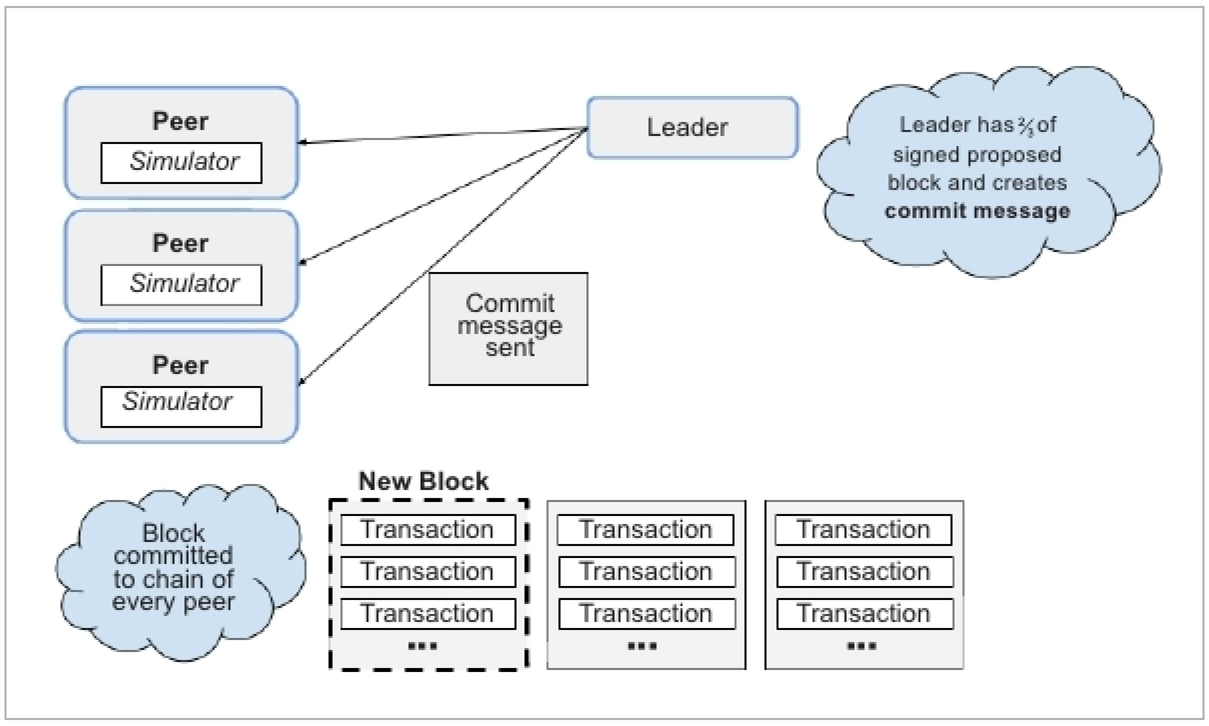

Figure 7: Transaction flow: Step 6

Step 2: After the peer performs stateless validation, the transaction is first sent to the ordering gate, which is responsible for choosing the right strategy for connection to the ordering service.

Step 3: The ordering service puts transactions into order and forwards them to peers in the consensus network in the form of proposals. A proposal is an unsigned block shared by the ordering service that contains a batch of ordered transactions. Proposals are only forwarded when the ordering service has accumulated enough transactions, or a certain amount of time has elapsed since the last proposal. This prevents the ordering service from sending empty proposals.

Step 4: Each peer verifies the proposal's contents (stateful validation) in the simulator, and creates a block that consists only of verified transactions. This block is then sent to the consensus gate, which performs YAC consensus logic.

Step 5: An ordered list of peers is determined, and a leader is elected based on the YAC consensus logic. Each peer casts a vote by signing and sending its proposed block to the leader.

Step 6: If the leader receives enough signed proposed blocks (i.e., more than two-thirds of the peers), then it starts to send a commit message, indicating that this block should be applied to the chain of each peer participating in the consensus. Once the commit message has been sent, the proposed block becomes the next block in the chain of every peer, via the synchroniser.

The YA Consensus algorithm

Hyperledger Iroha currently implements the YAC consensus algorithm, which is based on voting for block hash. The YAC performs two functions: ordering and consensus.

'Ordering' is responsible for ordering all transactions, packaging them into proposals, and sending them to peers in the network. The ordering service is an endpoint for setting an order of transactions and their broadcast (in a form of proposal). Ordering is not responsible for performing stateful validation of transactions.

'Consensus' involves taking blocks after they have been validated, collaborating with other blocks to decide on a commit, and propagating commits between peers. 'Consensus' is responsible for an agreement on blocks based on the same proposal.

'Validation' is an important part of the transaction flow; however, it is separate from the consensus process.

Note: Currently, the ordering service is a single point of failure that does the ordering and, therefore, Hyperledger Iroha is neither crash fault-tolerant, nor Byzantine fault-tolerant.

Hyperledger Iroha's mobile libraries

A major goal for the creators of Hyperledger Iroha is to develop a distributed ledger system that can be easily used by applications. In order to accomplish this, Hyperledger Iroha offers open source software libraries for iOS, Android, and JavaScript. These libraries allow for simple compatibility with not only Hyperledger Iroha, but also, potentially, with other networks through flexible API functions. These libraries are all open source, and available on GitHub.

An overview of the community

Hyperledger Iroha is an open source project for which ideas and code can be publicly discussed, created and reviewed. There are many ways to get involved with the Hyperledger Iroha community, one of which is through GitHub. END

References

[1] *https://github.com/hyperledger/iroha*
[2] *https://github.com/hyperledger/iroha-ios*
[3] *https://github.com/hyperledger/iroha-android*
[4] *https://github.com/hyperledger/iroha-javascript*
[5] *https://github.com/hyperledger/iroha-python*
[6] *https://github.com/hyperledger/iroha-scala*
[7] *https://github.com/hyperledger/iroha-dotnet*
[8] *https://github.com/hyperledger/iroha-api*

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